CISS240: Introduction to Programming
Quiz q1904

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. The following program does not run. Insert one statement to correct the problem.
Answer:

```
#include <iostream>

// insert statement here
int sum(int start, int end, int step);

int main()
{
    std::cout << sum(5, 10, 1) << std::endl;
    return 0;
}

int sum(int start, int end, int step)
{
    int s = 0;
    for (int i = start; i <= end; i += step)
    {
        s += i;
    }
    return s;
}
```

Q2. The following program does not run. Insert one statement to correct the problem.
Answer:

```
#include <iostream>
// insert statement here
void swap(int &x, int &y);

int main()
{
    int i, j;
    std::cin >> i >> j;
    swap(i, j);
    std::cout << i << ' ' << j << '\n';
    return 0;
}
```

```
void swap(int &x, int &y)
{
    int t = x;
    x = y;
    y = t;
}
```

Q3. The function `play_audio()` plays an audio file. The parameters are:

- `filename`: a C-string that is the name of the audio file to be loaded.
- `track_number`: an integer. 0 is the first track, 1 is the second track, etc.
- `repeat`: a boolean.

The function returns 0 if there are no errors, returns -1 if the file cannot be found using the `filename`, and returns -2 if the `track_number` is not found. Write down the prototype of this function.
Answer:

```
int play_audio(const char* filename, int track_number, bool repeat);
```

Q4. The function `copy_array()` copies the values of an array to another. The parameters are:

- `x`: the target integer array.
- `x_length`: an integer reference variable.
- `y`: the source integer array
- `y_length`: an integer variable.

When you call `copy_array(x, x_length, y, y_length)` the values `y[0]`,...,`y[y_length - 1]` are copied to `x[0]`,...,`x[y_length - 1]` (respectively) and `x_length` is set to `y_length`. Write down the prototype of this function.
Answer:

```
void copy_array(int x[], int &x_length, const int y[], int y_length);
```

Q5. What is a suitable prototype for `array_equals()` that compares the values of two integer arrays `x` and `y` from returns `true` when `x[0]`,...,`x[n - 1]` and `y[0]`,...,`y[n - 1]` are the same (respectively), and `false` otherwise.
Answer:

```
bool array_equals(const int x[], const int y[], int n);
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.